Homogenous equation বোঝার উপায়

(i) $(x^{2}+y^{2})dx - y^{2}dy = 0$ [power same]

(ii) $(x^{2}+xy)dx - xy\,dy = 0$ [গুণ থাকলে যোগ করে, power check]

(iii) $(y+\sqrt{x^2+y^2})dx - x\,dy = 0$ → [power same]

$(x^2+y^2)^{1/2}$

$x^{2\times 1/2}+y^{2\times 1/2}$

$x^{1}+y^{1}$

(iv) $v^{3}du - (u^{3} - uv^{2})dv = 0$ = power same

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step-1

→ $\frac{dy}{dx}$ form আনা, eqn (i)

step-2

$v = \frac{y}{x}$

$\Rightarrow \boxed{y = vx}$

Differentiating with x,

$\Rightarrow \frac{dy}{dx} = v.\frac{dx}{dx} + x\frac{dv}{dx}$

$\boxed{\Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}}$

y & $\frac{dy}{dx}$ eqn (i) এ বসানো।

Step 3: নতুন eqn থেকে dv সব v = dx সব x form বানানো

step 4:

→ integration করা,

→ V এর মান বসানো

→ c বের করা eqn (ii)

Step 5

∴ $y(1) = 6$

$\Rightarrow y(x) = y$

x, y দেয়া থাকলে c এর মান বের করে eqn (ii) তে বসানো।

1 $(y+\sqrt{x^2+y^2})\,dx - x\,dy = 0$, $y(1)=0$

$\therefore y+\sqrt{x^2+y^2}\,dx = x\,dy$

$\Rightarrow \frac{dy}{dx} = \frac{y+\sqrt{x^2+y^2}}{x} \longrightarrow (1)$

$v = \frac{y}{x}$

$\Rightarrow \boxed{y = vx}$

$\boxed{\Rightarrow \frac{dy}{dx} = v.1 + x\frac{dv}{dx}}$ put y & $\frac{dy}{dx}$ in eqn (1)

$\therefore v + x\frac{dv}{dx} = \frac{vx+\sqrt{x^2+v^2x^2}}{x}$

$\Rightarrow \quad " \quad = \frac{vx+\sqrt{x^2(1+v^2)}}{x}$

$\Rightarrow \quad " \quad = \frac{vx+x\sqrt{(1+v^2)}}{x}$

$\Rightarrow \quad " \quad = \frac{\cancel{x}(v+\sqrt{(1+v^2)})}{\cancel{x}}$

$\Rightarrow x\frac{dv}{dx} = \cancel{v}+\sqrt{(1+v^2)} - \cancel{v}$

$\Rightarrow x\frac{dv}{dx} = \sqrt{1+v^2}$

$\Rightarrow x\,dv = dx\,\sqrt{1+v^2}$

$\boxed{\Rightarrow dv\frac{1}{\sqrt{1+v^2}} = dx\,\frac{1}{x}}$

$\Rightarrow \int \frac{1}{\sqrt{1+v^2}}\,dv - \int \frac{1}{x}\,dx = 0$

$\left[\int \frac{1}{\sqrt{a^2+x^2}} = \ln\left[x+\sqrt{a^2+x^2}\right]\right]$

$\therefore \ln\left[v+\sqrt{1+v^2}\right] - \ln x = \ln c$

$\Rightarrow \ln\left[\frac{v+\sqrt{1+v^2}}{x}\right] = \ln c$

$\Rightarrow \frac{\frac{y}{x}+\sqrt{1+\frac{y^2}{x^2}}}{x} = c$

$\Rightarrow \frac{\frac{y}{x}+\sqrt{\frac{x^2+y^2}{x^2}}}{x} = c$

$\Rightarrow \frac{\frac{y}{x}+\frac{\sqrt{x^2+y^2}}{x}}{x} = c$

$\Rightarrow \frac{\frac{y+\sqrt{x^2+y^2}}{x}}{x} = c$

$\Rightarrow \frac{y+\sqrt{x^2+y^2}}{x^2} = c \longrightarrow (2)$

Given, $x=1$, $y=0$ put in eqn (2)

$\therefore \frac{0+\sqrt{1^2+0^2}}{1^2} = c$

$\Rightarrow c = 1$

$\therefore \boxed{y+\sqrt{x^2+y^2} = x^2}$ (Ans)

(2) $(x^2-3y^2)\,dx + 2xy\,dy = 0$

$\Rightarrow (x^2-3y^2)\,dx = -2xy\,dy$

$\Rightarrow \dfrac{dy}{dx} = \dfrac{x^2-3y^2}{-2xy}$

$\Rightarrow \dfrac{dy}{dx} = \dfrac{3y^2-x^2}{2xy} \longrightarrow (1)$

$V = \dfrac{y}{x}$

$\Rightarrow \boxed{y = Vx}$

$\Rightarrow \boxed{\dfrac{dy}{dx} = V + x\dfrac{dv}{dx}}$ in eqn (1)

$\therefore V + x\dfrac{dv}{dx} = \dfrac{3V^2x^2 - x^2}{2x^2V}$

$\Rightarrow V + x\dfrac{dv}{dx} = \dfrac{\cancel{x^2}(3V^2-1)}{2\cancel{x^2}V}$

$\Rightarrow x\dfrac{dv}{dx} = \dfrac{3V^2-1}{2V} - V$

$\Rightarrow x\dfrac{dv}{dx} = \dfrac{3V^2-1-2V^2}{2V}$

$\Rightarrow x\dfrac{dv}{dx} = \dfrac{V^2-1}{2V}$

$\boxed{\Rightarrow dv\,\dfrac{2V}{V^2-1} = dx\,\dfrac{1}{x}}$

$\Rightarrow \int \dfrac{2V}{V^2-1}\,dv - \int \dfrac{1}{x}\,dx = 0$

$\left[\int \dfrac{f'(x)}{f(x)} = \ln[f(x)]\right]$

$\ln[V^2-1] - \ln[x] = \ln c$

$\Rightarrow \ln\left[\dfrac{V^2-1}{x}\right] = \ln c$

$\Rightarrow \dfrac{V^2-1}{x} = c \longrightarrow (2)$

$\therefore \dfrac{\frac{y^2}{x^2} - 1}{x} = c$

$\Rightarrow \dfrac{\frac{y^2-x^2}{x^2}}{x} = c$

$\Rightarrow \boxed{\dfrac{y^2-x^2}{x^3} = c}$ (Ans)

③ $v^3 du - (u^3 - uv^2) dv = 0$

$\boxed{\cancel{\Rightarrow \frac{dv}{du} = \frac{v^3}{u^3 - uv^2}} \rightarrow ①}$ $\frac{du}{dv} = \frac{u^3 - uv^2}{v^3}$

$\boxed{\cancel{\Rightarrow v = vx} \quad \cancel{\Rightarrow y = x + x\frac{dv}{dx}}}$ $z = \frac{u}{v}$

$\Rightarrow u = zv$

$\Rightarrow \frac{du}{dv} = z + v\frac{dz}{dv}$

$\therefore z + v\frac{dz}{dv} = \frac{z^3v^3 - zv^3}{v^3}$

$\Rightarrow v\frac{dz}{dv} = z^3 - z - z$

$\Rightarrow \frac{1}{z^3 - 2z}dz - dv\frac{1}{v} = 0$

$\Rightarrow \int \frac{1}{z(z^2-2)}dz - \int \frac{1}{v}dv = 0$

Partial;

$\therefore \int \frac{1}{z(z+\sqrt{2})(z-\sqrt{2})} = \frac{A}{z} + \frac{B}{z+\sqrt{2}} + \frac{C}{z-\sqrt{2}}$

$\Rightarrow 1 = A(z+\sqrt{2})(z-\sqrt{2}) + B\{z(z-\sqrt{2})\} + C\{z(z+\sqrt{2})\}$

$\Rightarrow 1 = A(z^2-2) + B(z^2 - z\sqrt{2}) + C(z^2 + \sqrt{2})$

$z = 0$	$z = +\sqrt{2}$	$z = -\sqrt{2}$
$\Rightarrow 1 = A(0-2) + 0 + 0$	$\Rightarrow 1 = 0 + 0 + 4C$	$\Rightarrow 1 = 0 + 4B + 0$
$\Rightarrow A = -\frac{1}{2}$	$\Rightarrow C = \frac{1}{4}$	$\Rightarrow B = \frac{1}{4}$

$\Rightarrow \int \text{———} = -\frac{1}{2z} + \frac{1}{4(z+\sqrt{2})} + \frac{1}{4(z-\sqrt{2})}$ △

$\Rightarrow -\frac{1}{2}\int \frac{1}{z}dz + \frac{1}{4}\int \frac{1}{z+\sqrt{2}}dz + \frac{1}{4}\int \frac{1}{z+\sqrt{2}}dz$

$\Rightarrow -\frac{1}{2}\ln(z) + \frac{1}{4}\ln(z+\sqrt{2}) + \frac{1}{4}\ln(z+\sqrt{2})$

$\Rightarrow \left(-\frac{1}{2}\ln(z) + \frac{1}{4}\ln(z^2-2) - \ln(v) = 0\right) \times 4$

$\Rightarrow -2\ln(z) + \ln(z^2-2) - 4\ln(v) = 0$

$\Rightarrow -\ln(z)^2 + \ln(z^2-2) - \ln(v)^4 = \ln C$

$\Rightarrow \frac{z^2-2}{z^2 \times v^4} = C$

$\Rightarrow \frac{\frac{u^2}{v^2} - 2}{\frac{u^2}{v^2} \times v^4} = C$

$\Rightarrow \frac{\frac{u^2 - 2v^2}{v^2}}{u^2v^2} = C$

$\Rightarrow u^2 - 2v^2 = cu^2v^4$

4 $(2xy+3y^2)dx-(2xy+x^2)dy=0$

$$\Rightarrow \frac{dy}{dx}=\frac{2xy+3y^2}{2xy+x^2} \rightarrow ①$$

$$\therefore \frac{dy}{dx}=v+x\frac{dv}{dx} \qquad \because y=vx$$

$$\therefore v+x\frac{dv}{dx}=\frac{2x^2v+3v^2x^2}{2x^2v+x^2}$$

$$\Rightarrow v+x\frac{dv}{dx}=\frac{x^2(2v+3v^2)}{x^2(2v+1)}$$

$$\Rightarrow x\frac{dv}{dx}=\frac{2v+3v^2}{2v+1}-v$$

$$\Rightarrow x\frac{dv}{dx}=\frac{2v+3v^2-2v^2-v}{2v+1}$$

$$\Rightarrow x\frac{dv}{dx}=\frac{v^2+v}{2v+1}$$

$$\boxed{\Rightarrow dv\frac{2v+1}{v^2+v}=dx\frac{1}{x}}$$

$$\Rightarrow \int\frac{2v+1}{v^2+v}dv-\int\frac{1}{x}dx=0$$

$$\Rightarrow \ln[v^2+v]-\ln[x]=\ln c$$

$$\Rightarrow \ln\left[\frac{v^2+v}{x}\right]=\ln c$$

$$\Rightarrow \frac{v^2+v}{x}=c$$

$$\Rightarrow \frac{\frac{y^2}{x^2}+\frac{y}{x}}{x}=c$$

$$\Rightarrow \frac{\frac{y^2+xy}{x^2}}{x}=c$$

$$\boxed{\Rightarrow \frac{y^2+xy}{x^3}=c}$$ (Ans)

5 $(x^2+3y^2)dx-2xy\,dy=0,\ y(2)=6$

$$\Rightarrow \frac{dy}{dx}=\frac{x^2+3y^2}{2xy} \rightarrow ①$$

$$\frac{dy}{dx}=v+x\frac{dv}{dx} \qquad \because y=vx$$

$$\therefore v+x\frac{dv}{dx}=\frac{x^2+3x^2v^2}{2vx^2}$$

$$\Rightarrow v+x\frac{dv}{dx}=\frac{x^2(1+3v^2)}{x^2 2v}$$

$$\Rightarrow x\frac{dv}{dx}=\frac{1+3v^2}{2v}-v$$

$$\Rightarrow x\frac{dv}{dx}=\frac{1+3v^2-2v^2}{2v}$$

$$\Rightarrow x\frac{dv}{dx}=\frac{1+v^2}{2v}$$

$$\boxed{\Rightarrow dv\frac{2v}{1+v^2}=dx\frac{1}{x}}$$

$$\Rightarrow \int\frac{2v}{1+v^2}dv-\int\frac{1}{x}dx=0 \qquad \left[\int\frac{f'(x)}{f(x)}=\ln f(x)\right]$$

$$\Rightarrow \ln(1+v^2)-\ln(x)=\ln(c)$$

$$\Rightarrow \ln\left(\frac{1+v^2}{x}\right)=\ln c$$

$$\Rightarrow \frac{1+\frac{y^2}{x^2}}{x}=c$$

$$\Rightarrow \frac{\frac{x^2+y^2}{x^2}}{x}=c$$

$$\boxed{\Rightarrow \frac{x^2+y^2}{x^3}=c}$$

$$\therefore y(2)=6 \quad (x,\ y)$$

$$\therefore \frac{4+36}{8}=c$$

$$\Rightarrow \boxed{c=5} \rightarrow \boxed{x^2+y^2=5x^3}$$ (Ans)

⑥ $(2x-5y)\,dx + (4x-y)\,dy = 0$, $y(1) = 4$

$\Rightarrow \frac{dy}{dx} = \frac{-(2x-5y)}{4x-y}$

$\Rightarrow \frac{dy}{dx} = \frac{5y+2x}{4x-y}$

$v = y/x$

$y = vx$

$\frac{dy}{dx} = v + x\frac{dv}{dx}$

$\therefore v + x\frac{dv}{dx} = \frac{5vx \mp 2x}{4x - vx}$

$\Rightarrow v + x\frac{dv}{dx} = \frac{5v \mp 2}{4-v}$

$\Rightarrow x\frac{dv}{dx} = \frac{5v \mp 2}{4-v} - v$

$\Rightarrow x\frac{dv}{dx} = \frac{5v \mp 2 - (4v - v^2)}{4-v}$

$\Rightarrow x\frac{dv}{dx} = \frac{5v \mp 2 - 4v + v^2}{4-v}$

$\Rightarrow x\frac{dv}{dx} = \frac{v^2 + v \mp 2}{4-v}$

$\Rightarrow dv\,\frac{4-v}{v^2+v\mp 2} = dx\,\frac{1}{x}$

$\Rightarrow \int \frac{4-v}{v^2+v\mp 2}\,dv - \int \frac{1}{x}\,dx = 0$

Partial

$\int \frac{4-v}{(v+2)(v-1)} \equiv \frac{A}{v+2} + \frac{B}{v-1}$

$\Rightarrow 4 - v = A(v-1) + B(v+2)$

$\therefore v - 1 = 0$

$\Rightarrow v = 1$

$\therefore 4 - 1 = A(1-1) + B(1+2)$

$\Rightarrow 3 = 3B$

$\Rightarrow B = 1$

$\therefore v + 2 = 0$

$\Rightarrow v = -2$

$\therefore 4 - (-2) = A(-2-1) + 0$

$\Rightarrow 6 = -3A$

$\Rightarrow A = -2$

$\int \frac{4-v}{(v+2)(v-1)} = \int -\frac{2}{v+2} + \int \frac{1}{v-1}\,dv$

$= -2\ln(v+2) + \ln(v-1)$

$= \ln(v+2)^{-2} + \ln(v-1)$

$= \ln\left\{\frac{v-1}{(v+2)^2}\right\}$

$\Rightarrow \ln\left\{\frac{v-1}{(v+2)^2}\right\} - \ln x = \ln c$

$\Rightarrow \frac{v-1}{x(v+2)^2} = c$

$\Rightarrow \frac{\frac{y}{x} - 1}{x\left(\frac{y}{x}+2\right)^2} = c$

$\Rightarrow \frac{\frac{y-x}{x}}{x\left(\frac{y}{x}+2\right)^2} = c$

$\Rightarrow \frac{\frac{y-x}{x}}{x\left(\frac{y+2x}{x}\right)^2} = c$

$\Rightarrow \frac{\frac{y-x}{\cancel{x}}}{\frac{(y+2x)^2}{\cancel{x}}} = c \Rightarrow \boxed{\frac{y-x}{(y+2x)^2} = c}$ (Ans)

$y(1) = 4$
$\quad\; x \quad\; y$

$\therefore \frac{4-1}{(4+2)^2} = c$

$\Rightarrow \frac{3}{36} = c$

$\Rightarrow \frac{1}{12} = c$

$\Rightarrow \boxed{\frac{y-x}{(y+2x)^2} = \frac{1}{12}}$ (Ans)